22-23秋线性代数期中(回忆版)

1. Multiple Choice. Only one choice is correct.

(1) Which of the following statements can guarantee that the following homogeneous system of linear equations $\begin{cases} a_{11}x_1+a_{12}x_2+\cdots+a_{1n}x_n=0 \\ \cdots\cdots\cdots\cdots \\ a_{m1}x_1+a_{m2}x_2+\cdots+a_{mn}x_n=0 \end{cases}$ has a nonzero solution?

(A) $m\leq n$ (B) $m=n$ (C) $m>n$ (D) The rank of the coefficient matrix is less than n.

(2) Which of the following matrices can be written as a product of elementary matrices?

(A) $\begin{bmatrix}1&2&3\\0&4&2\end{bmatrix}$ (B) $\begin{bmatrix}1&2&0\\0&-1&3\\0&0&2\end{bmatrix}$ (C) $\begin{bmatrix}1&0&1\\1&0&1\\0&1&0\end{bmatrix}$ (D) $\begin{bmatrix}-3&2&7\\-1&2&3\\0&-2&-1\end{bmatrix}$

(3) Let β_1,β_2,β_3 be a basis of the null space $N(A)$ of some matrix A. Another basis of $N(A)$ is ( ).

(A) $\beta_1+\beta_2,\beta_2+\beta_3,\beta_3+\beta_1$ (B) $\beta_1+\beta_2,\beta_2+\beta_3,\beta_3-\beta_1$

(C) $\beta_1-\beta_2,\beta_2-\beta_3,\beta_3-\beta_1$ (D) $\beta_1+2\beta_2,2\beta_2+3\beta_3,3\beta_3-\beta_1$.

(4) Let $a,b\in R$. The set

$$V=\{(x,y,z,w): x+2y+3z+4w=a+b+1,\ x-2y+4z-w=a-2b-5\}$$

is a subspace of R^4 if

(A) $a=-1,b=1$ (B) $a=-2,b=1$ (C) $a=1,b=-2$. (D) $a=1,b=-1$.

(5) Let u and v be unit vectors in R^3. If the vectors $u+2v$ and $5u-4v$ are orthogonal, then the angle α between u and v is ( ).

(A) $\alpha=\frac{\pi}{6}$ (B) $\alpha=\frac{\pi}{4}$ (C) $\alpha=\frac{\pi}{3}$ (D) $\alpha=\frac{3\pi}{4}$

2. Fill in the blanks.

(1) Let A be a 5×8 real matrix. If $\dim N(A)=3$, then $\dim(N(A^T))=$ _____.

(2) All the 2×2 matrices that commute with $\begin{bmatrix}1&0\\1&1\end{bmatrix}$ can be written in the form _____.

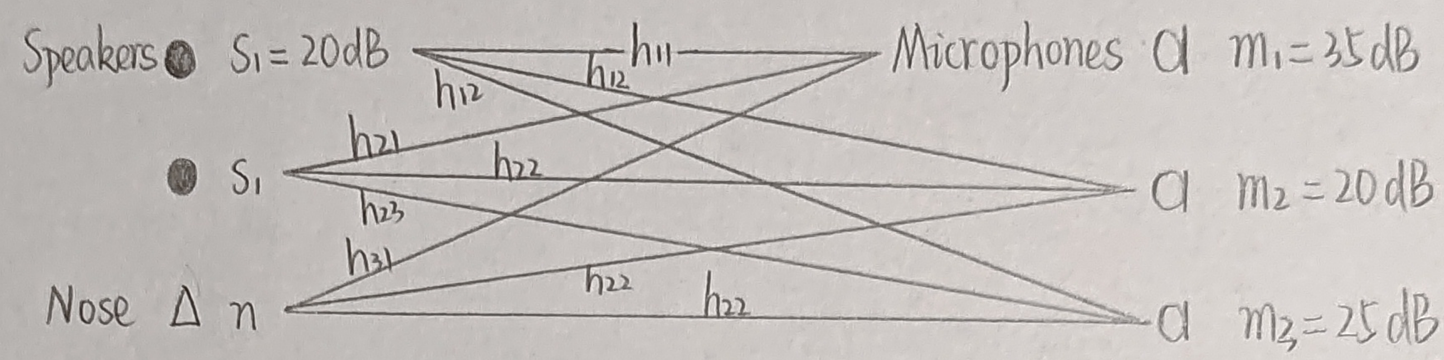
(3) Let $A=\begin{bmatrix}1&2&-2\\2&-1&\lambda\\3&1&-1\end{bmatrix}$. If $AB=0$ for some nonzero matrix B, then $\lambda=$ _____.

(4) An LU-factorization of $A=\begin{bmatrix}2&1\\3&7\end{bmatrix}$ is $L=$ _____, $U=$ _____.

(5) Suppose b is a nonzero column vector. If η_1,η_2 are solutions to the system of linear equations $Ax=b$, and $\lambda_1\eta_1+\lambda_2\eta_2$ is another solution to $Ax=b$, then λ_1,λ_2 must satisfy _____.

3. An audio processing company develops technology for mobile devies and is proud of the capacity of their products to filter surrounding noise. Here is a simplified model (a single-

layer neural network) showing how it works. Let s_1, s_2 be the volumes of a pair of speakers and n denote that of noise. Use 3 microphones to receive ~~single~~ signals with recorded volumes m_1, m_2 and m_3. All values are in decribels and shown in the following diagram, where the linear factor h_{ij} indicates the rate of decay along each channel. (When a sound of 100 dB is transmitted along a channel with rate of decay h, the volume received is $100h$ dB.)



Suppose we are given the matrix $[h_{ij}] = \begin{bmatrix} 0.875 & 0.5 & 0.75 \\ 0.25 & 0.5 & 0.5 \\ 0.625 & 0.375 & 0.5 \end{bmatrix}$. Estimate the volume of the unknown sourse speaker by solving a linear system for s_2.

4. Let T be a linear transformation from R^3 to R^3 such that

$$T(\alpha_1) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, T(\alpha_2) = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, T(\alpha_3) = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \text{ where } \alpha_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}.$$

(a) Show that $\alpha_1, \alpha_2, \alpha_3$ is a basis of R^3.

(b) Find the representation matrix A of T (in the standard basis e_1, e_2, e_3 of R^3)

(c) Is the matrix A invertible? Why?

5. Let L be the line of intersection of $x_1 + x_2 + x_3 = 0$ and $2x_1 - x_2 - 2x_3 = 0$ in R^3.
Find the orthogonal projection of $b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ onto L.

6. Let u, v be nonzero column vectors in R^n and $A = uv^T$.

(a) Prove that the rank of A is 1.

(b) What are the possible values of the rank of the matrix $\begin{bmatrix} u^Tv & 0 \\ 0 & vu^T \end{bmatrix}$? Justify your answer.

7. Let $\alpha_1, \cdots, \alpha_n$ be column vectors in R^3. Suppose that the systerm $\alpha_1, \alpha_2, \cdots, \alpha_{n-1}$ is linearly dependent, and that the system $\alpha_2, \alpha_3, \cdots, \alpha_n$ is linearly independent.
Let $A = (\alpha_1, \alpha_2, \alpha_3, \cdots, \alpha_n)$ and $\beta = \alpha_1 + \alpha_2 + \cdots + \alpha_n$.

(a) Show that α_1 can be written as a linear ~~system~~ combination of $\alpha_2, \alpha_3, \cdots, \alpha_n$, i.e, there exist contants $k_2, k_3, \cdots, k_n$ so that $\alpha_1 = k_2\alpha_2 + k_3\alpha_3 + \cdots + k_n\alpha_n$.

(b) Show that the linear system $Ax = \beta$ has infinitely many solutions.

(c) Prove that if $n > 2$, then $A^2 \neq 0$. Here 0 denotes the zero matrix of order n.